Insight46: Comparison between Freesurfer and GIF for amyloid PET quantification

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This document provides some basic information about the processing done to the PET and MR images to obtain amyloid PET SUVR values using Freesurfer-based brain parcellations, and how these outputs compare to the primary analysis technique we have used in the Insight46 study, which parcellates the MRI using our in-house Geodesic Information FLows (GIF) pipeline. It should serve as an accompaniment to the spreadsheet and data dictionary that has the Amyloid PET FreeSurfer values.

Which set of SUVR values should I use

There have been multiple approaches tried and tested with the Insight46 dataset. we have used the GIF method for a large amount of in-house analyses. Of the different flavors tested, we have found the SUVR pipeline using the whole cerebellum as a reference region, with partial volume correction applied, provides the best results over a number of factors, and this is the current method that we have selected for the main analyses within the study. Further information can be found in (Coath et al. 2026). However, there have been multiple requests to include the Insight46 data into larger data sets, and for ease of harmonisation of this data with other work, we would suggest using FreeSurfer with the whole cerebellum.

To summarise-

Use **GIF SUVR** if:

- You analysis only involves data from Insight46.

Use **FreeSurfer SUVR** if:

- You plan to merge this dataset with other available data sets, where a FreeSurfer based pipeline has likely been used to process the data.

Methods

Full details around the study, image acquisition, and image processing pipelines can be found in numerous Insight46 studies, including Coath et al. (2023)

MRI processing

The preferred structural T1 scan from each imaging session was used. For each session, two brain parcellation techniques were used. The first method was Geodesic Information Flows (Cardoso et al. 2015), which is a multi-atlas based segmentation propogation algorithm that simultaneously perfoms bias correction, tissue segmentation, and brain parcellation using the Neuromorphometrics atlas. The second approach was running Freesurfer 7.1 (Fischl 2012) on the T1 scan, using the cortical parcellations from the Desikan-Killiany atlas.

For each parcellation, ROI's were combined into six lobar regions: frontal, temporal, occipital, cingulate, parietal, and insula. Left and right lobes were averaged together (weighted by volume) to get bilateral SUVRs. Two regions, the posterior cingulate and the precuenus, are not normally included in the lobar values for GIF, as they have been identified as sites where early amyloid accumulation can be detected, and so they

were included as individual regions. For comparison with Freesurfer-based PET pipelines in other studies, we have chosen to keep the precuenus and posterior cingualte as part of the parietal and cingulate loves respectively, so these values could better match with future analyses and meta-analyses using the FreeSurfer values. Therefore, for this comparison only, we have also added these regions into the GIF lobar SUVR to have a better comparison with the FS lobar values. You can find the individual ROIs that make up each lobe for both FreeSurfer and GIF in the Appendix

A global composite region was also created, for Freesurfer this is based on the ROI's found in the ADNI composite, which can be found in the ADNI Florbetapir briefing document. A composite using the GIF regions was based on combining the regions that provided the closest match to the FreeSurfer composite.

PET processing

All PET processing was acquired and performed in the same manner. Late-frame amyloid PET (50-60 minutes post injection) using florbetapir was co-registered to the MRI using NiftyPet (Modat et al. 2014) and the parcellated ROIs were transferred to the PET using the transformation from the co-registration. The whole cerebellum was chosen as the reference region. Standardized Uptake Value Ratios (SUVR) were obtained by dividing the mean activity in the target region of interest by the mean activity in the reference region.

Analysis

This document describes the comparison between SUVR's obtained from the GIF pipeline and those using the FreeSurfer pipeline. As the anatomical parcellation scheme is different, we are only comparing at the lobar and global composite level. We want to look at:

- The association of SUVR between parcellations
- The difference in cutpoints and amyloid positivity rates
- The difference in longitudinal rates.

While we already have established a Centiloid calibration for our GIF based pipelines (Coath et al. 2023), we have not yet done the calibration for the Freesurfer pipeline so comparison of Centiloids is not yet possible. Other reference regions (eroded white matter, cerebellar grey matter, pons, composie reference region) are also available for the FreeSurfer and will be released subesequently if requested.

Results

Association between FS and GIF SUVR

First a linear regression was performed between the FreeSurfer and GIF SUVR values for each lobe. **Table 1** provides the results from these regression. The results suggest that there is a strong association between the FS and GIF SUVR values, with all lobar regions having R^2 values above 0.95. Most of the linear relationships between FS and GIF are near the unity line, with intercepts near zero and slopes near 1. Subcortical regions, of which there is limited uptake in early stages of AD, also showed high levels of agreement, with all structures but the thalamus having an R^2 above 0.9. In all of the lobes except the temporal region, the intercepts are significantly different than 0 and show a small positive value, indicating a slight offset of increased SUVR in FreeSurfer compared to GIF. The insual region has the lowest R^2 values, and deviates the most from the unity line in terms of both slope and intercept.

Table 1: Slope, intercept and R-squared values for Freesurfer/GIF SUVR relationships

	Intercept			Slope			
ROI	Cofficient	Std Error	p	Cofficient	Std Error	p	R^2
frontal	0.016	0.005	<0.001	1.009	0.004	<0.001	0.985

Table 1: Slope, intercept and R-squared values for Freesurfer/GIF SUVR relationships

	Intercept			Slope			
ROI	Cofficient	Std Error	p	Cofficient	Std Error	p	R^2
temporal	0.002	0.005	0.684	1.016	0.005	<0.001	0.984
parietal	0.044	0.004	<0.001	0.984	0.004	<0.001	0.986
occipital	0.050	0.007	<0.001	0.979	0.007	<0.001	0.962
cingulate	0.034	0.005	<0.001	0.985	0.004	<0.001	0.984
insula	0.147	0.007	<0.001	0.915	0.007	<0.001	0.958
composite	0.018	0.004	<0.001	0.998	0.004	<0.001	0.988
corticalgm	0.009	0.004	0.024	1.013	0.004	<0.001	0.988
hippocampus	0.017	0.008	0.037	0.996	0.008	<0.001	0.954
thalamus	0.041	0.019	0.030	1.054	0.017	<0.001	0.833
caudate	-0.014	0.010	0.164	1.027	0.010	<0.001	0.934
putamen	0.110	0.010	<0.001	0.971	0.008	<0.001	0.946
subcorticalgm	0.043	0.010	<0.001	1.017	0.009	<0.001	0.947

Figure 1 shows the linear relationship between GIF and FreeSurfer for each of these lobes, which highlights how closely aligned these SUVR values are, particularly for the global composite region.

Figure 2 shows the same relationship in a slightly different fashion, as Bland-Altman plots for each of these lobes, again demonstrating the tight agreement between SUVRs with a tendency for FreeSurfer to overestimate SUVR's compared to GIF. The slight positive relationship in the insula region with respect to mean signal likely reflects the lower slope observed in the linear regression for this lobe.

Amyloid positivity rates

Cutpoints for amyloid positivity were determined using a Gaussian Mixture model. Fits using one, two and three Gaussians were performed, with the best one selected based on Baysian Information Criteria. As in all of the work involving Insight46 Amyloid PET data, the cutpoint was selected to be the 99th percentile of the larger lower Gaussian distribution which represents the amyloid negative individuals.

Table 2 summarizes the GMM results for FreeSurfer and GIV, and provides the resulting cutoff values and positivity rates using these positivity rates and the cutoff values between the FS and GIF. In general the cutoff is higher in FreeSurfer, which is line with the correlations shown above. However, the positivity rates are still quite similar, usually 1-2% different. In some lobes, FS has slightly higher positivity rates and lower rates in others. Many subcortical structures show the best fit with one gaussian, suggesting that these structures do not have a bimodal distribution, and they are not useful for separating amyloid negative individuals from those who are amyloid positive. The main exception is the insula, which has noticeably higher positivity rates in GIF compared to FreeSurfer.

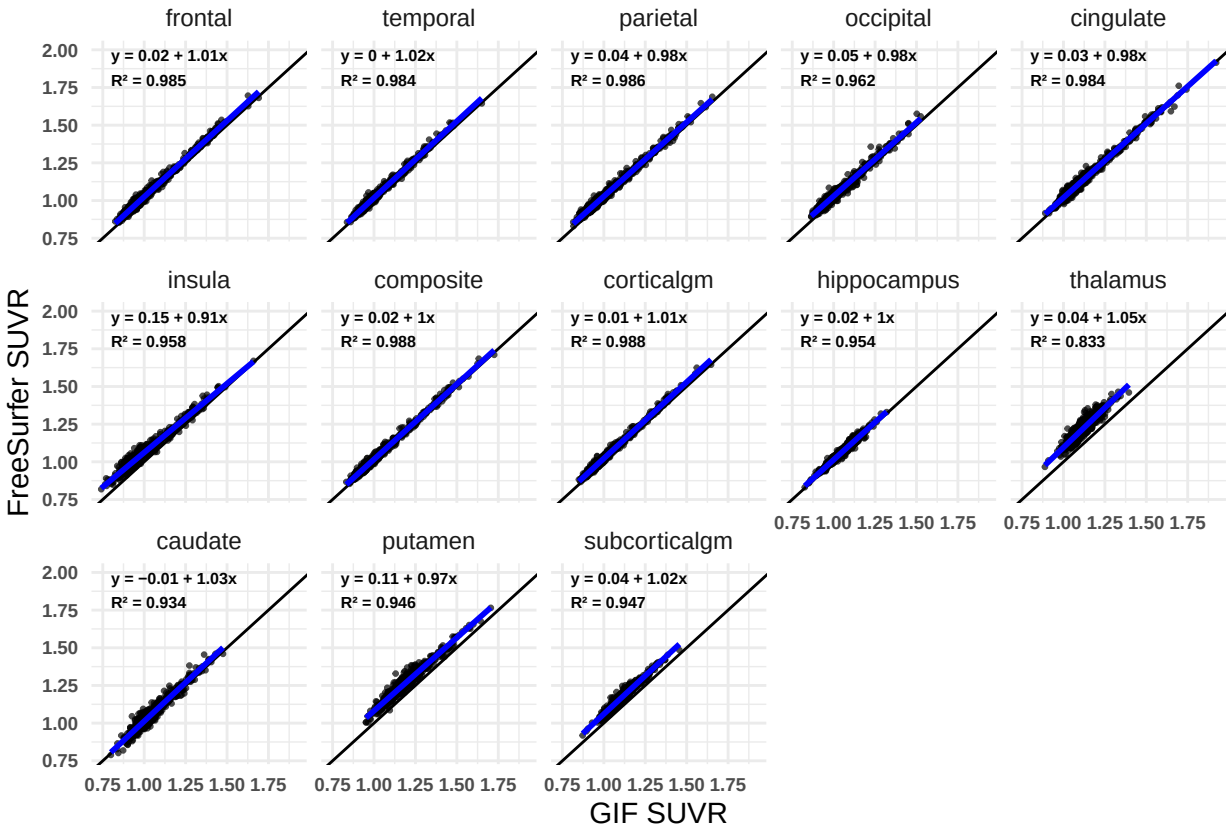


Figure 1: Correlation scatter plots of each lobe between FreeSurfer SUVR and GIF SUVR

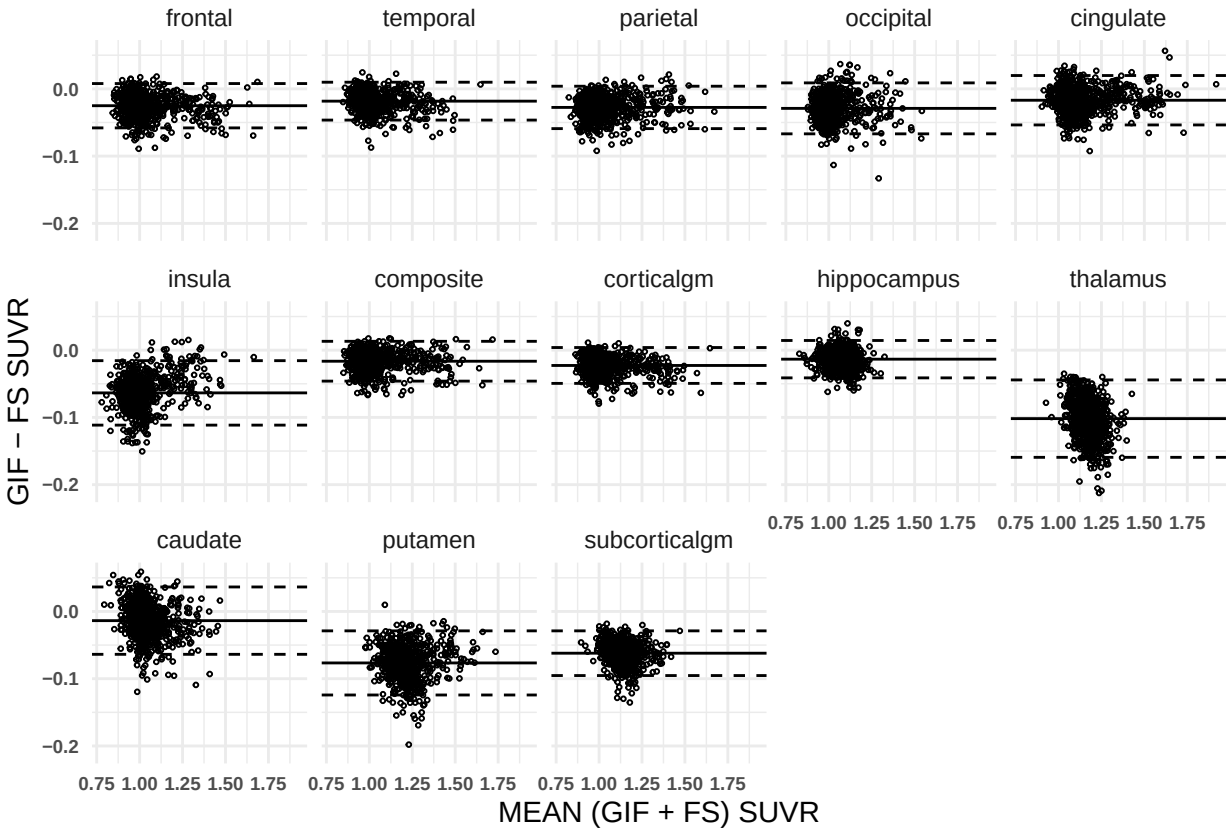


Figure 2: Bland-Altman plots per lobe comparing FreeSurfer SUVR and GIF SUVR

Table 2: GMM-based cutoffs and associated positivity rates by lobe for FreeSurfer and GIF

				Baseline	Followup
ROI	Pipeline	# Gaussians	Cutpoint	N positive (%)	N positive (%)
frontal	fs	2	1.093	79 (18.2)	93 (26.1)
	gif	2	1.066	81 (18.7)	92 (25.8)
temporal	fs	2	1.084	79 (18.2)	86 (24.2)
	gif	2	1.063	74 (17.1)	85 (23.9)
parietal	fs	2	1.089	91 (21.0)	89 (25.0)
	gif	2	1.067	86 (19.9)	85 (23.9)
occipital	fs	2	1.109	56 (12.9)	56 (15.7)
	gif	2	1.071	65 (15.0)	65 (18.3)
cingulate	fs	2	1.218	75 (17.3)	83 (23.3)
	gif	2	1.189	83 (19.2)	84 (23.6)
insula	fs	2	1.117	52 (12.0)	67 (18.8)
	gif	2	1.033	74 (17.1)	77 (21.6)
composite	fs	2	1.083	88 (20.3)	93 (26.1)
	gif	2	1.068	85 (19.6)	92 (25.8)
corticalgm	fs	2	1.100	73 (16.9)	79 (22.2)
	gif	2	1.074	79 (18.2)	83 (23.3)
hippocampus	fs	1	1.220	6 (1.4)	6 (1.7)
	gif	1	1.205	6 (1.4)	5 (1.4)
thalamus	fs	1	1.380	7 (1.6)	6 (1.7)
	gif	2	1.243	8 (1.8)	12 (3.4)
caudate	fs	2	1.183	41 (9.5)	33 (9.3)
	gif	2	1.116	61 (14.1)	60 (16.9)
putamen	fs	2	1.350	52 (12.0)	58 (16.3)
	gif	2	1.266	51 (11.8)	56 (15.7)
subcorticalgm	fs	2	1.288	22 (5.1)	25 (7.0)
	gif	2	1.183	46 (10.6)	46 (12.9)

Longitudinal change comparison

Finally, we compared the two methods for measuring longitudinal rates of change in amyloid accumulation between the baseline scan and the followup scan ~ 2.5 years later. Table 3 provides a comparison of the longitudinal change values between FreeSurfer and GIF.The R^2 values are lower in the longitudinal change measurements compared to cross-sectional measures, and this may simply reflect increased intra-subject variability (measurement error) in these regions. In amyloid negative individuals, where it is unlikely that

actual amyloid accumulation is taking place, this measurement error is likely the largest source of varation. In The intercepts are statistically significantly different from zero in all lobes but occipital and temporal, but these values are quite small. The slopes slopes remain close to 1.

Table 3: Slope, intercept and R-squared values for relationships in longitudinal change of Freesurfer/GIF SUVR

	Intercept			Slope			
ROI	Cofficient	Std Error	p	Cofficient	Std Error	p	R^2
delta_frontal	0.004	0.001	<0.001	1.018	0.018	<0.001	0.905
delta_temporal	0.000	0.001	0.661	1.027	0.019	<0.001	0.902
delta_parietal	0.004	0.001	<0.001	0.988	0.019	<0.001	0.898
delta_occipital	-0.001	0.001	0.198	0.965	0.028	<0.001	0.791
delta_cingulate	0.002	0.001	0.044	0.998	0.019	<0.001	0.892
delta_insula	0.003	0.001	0.026	0.978	0.029	<0.001	0.776
delta_composite	0.003	0.001	<0.001	1.018	0.014	<0.001	0.939
delta_cortical_gm	0.002	0.001	0.004	1.043	0.015	<0.001	0.937
delta_hippocampus	0.001	0.001	0.489	0.954	0.024	<0.001	0.826
delta_thalamus	0.002	0.002	0.260	1.103	0.046	<0.001	0.640
delta_caudate	-0.001	0.002	0.666	0.921	0.037	<0.001	0.661
delta_putamen	0.000	0.001	0.986	1.040	0.026	<0.001	0.827
delta_subcortical_gm	-0.001	0.001	0.452	1.129	0.027	<0.001	0.847

Figure 3 shows the scatter plot of these relationships for each lobe. As would be expected, very few points have negative slopes.

Exported files

All the FreeSurfer SUVR files have been exported and will be part of Insigh46's main data collection at Skylark.

Take home messages

- At a global composite and lobar level, FreeSurfer and GIF parcellations produced very similar results in both cross-sectional measures of uptake, and in changes over time.
- FreeSurfer SUVRs tend to be slightly higher than GIF. THis might suggest a difference in partial volume effects within the parcellation, with one method perhaps being a bit tighter to GM than the other.
- The insula region is where the most discrepancies are observed, and this may be due to different interpretations of the structures included in this lobe.

Appendix: ROI assignments in lobes

The following tables show which ROI's are assigned to each lobe in the SUVR process for both FreeSurfer and GIF.

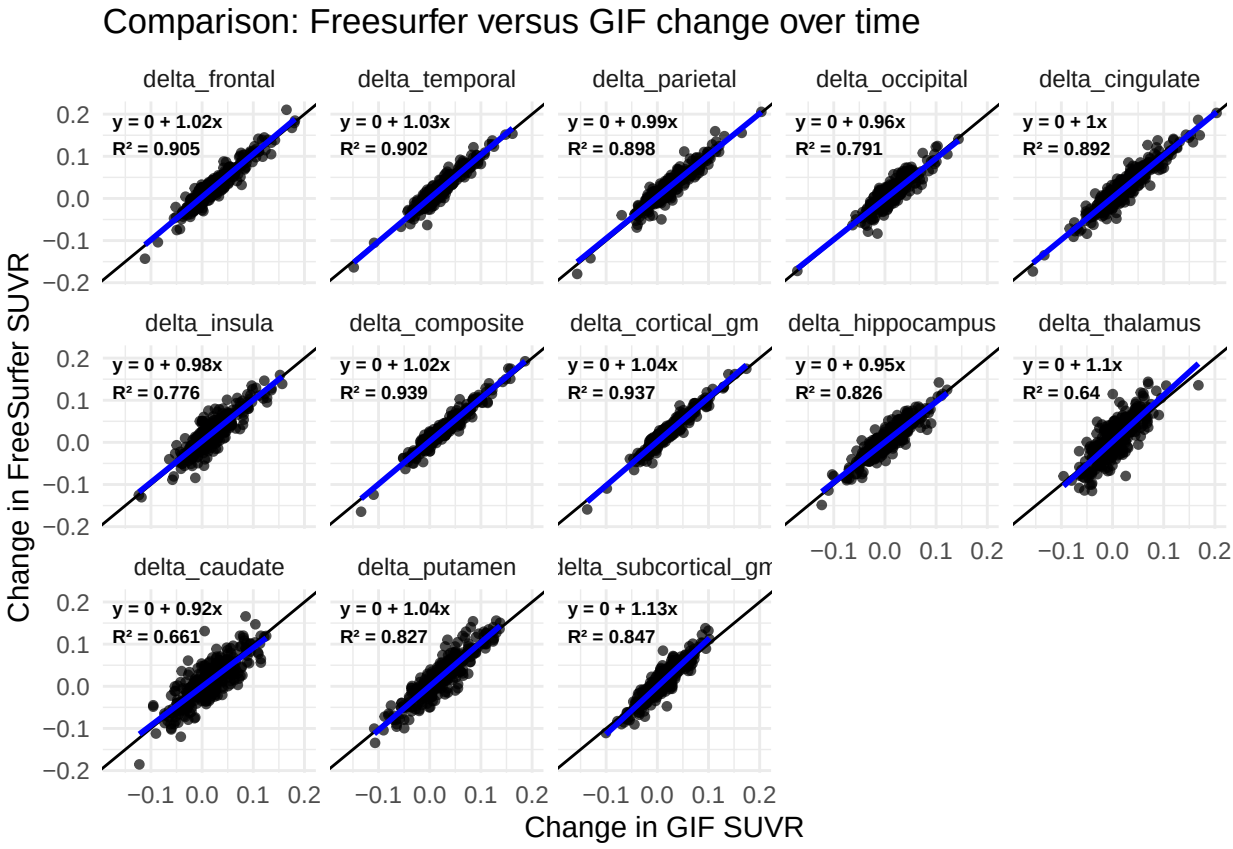


Figure 3: Correlation scatter plots of each lobe between longitudinal change in FreeSurfer SUVR and GIF SUVR

Table 4: Freesurfer: structures in Desikan-Killiany atlas by lobe, based on Freesurfer guidance (with L/R labels)

Frontal	Temporal	Occipital	Cingulate	Parietal	Insula
Superior Frontal (1028/2028)	Superior, Temporal (1030/2030)	Lateral Occipital (1011/2011)	Rostral Anterior (1026/2026)	Superior Parietal (1029/2029)	Insula
Rostral Middle Frontal (1027/2027)	Middle Temporal (1015/2015)	Lingual (1013/2013)	Caudal Anterior (1002/2002)	Inferior Parietal (1008/2008)	
Caudal Middle Frontal (1003/2003)	Inferior Temporal (1009/2009)	Cuneus (1005/2005)	Posterior (1023/2023)	Supramarginal (1031/2031)	
Pars Opercularis (1018/2018)	Banks of the Superior Temporal Sulcus (1001/2001)	Pericalcarine (1021/2021)	Isthmus (1010/2010)	Postcentral (1022/2022)	
Pars Orbitalis (1019/2019)	Fusiform (1007/2007)			Precuneus (1025/2025)	
Pars Triangularis (1020/2020)	Transverse Temporal (1034/2034)				
Lateral Orbitofrontal (1012/2012)	Entorhinal (1006/2006)				
Medial Orbitofrontal (1014/2014)	Temporal Pole (1033/2033)				
Precentral (1024/2024)	Parahippocampal (1016/2016)				
Paracentral (1017/2017)					
Frontal Pole (1032/2032)					

Table 5: GIF: cortical structures in Neuromorphometrics atlas by lobe, based on designation in GENFI study. THe italicised structure (posterior cingulate and precuneus) were structures that are normally not included in the lobar values for GIF but have been included here for comparison to FreeSurfer.

Frontal	Temporal	Occipital	Cingulate	Parietal	Insula
Anterior orbital gyrus (105/106)	Entorhinal area (117/118)	Calcarine cortex (109/110)	Anterior cingulate (101/102)	Angular gyrus (107/108)	Anterior insula (103/104)
Central operculum (113/114)	Fusiform gyrus (123/124)	Cuneus (115/116)	Middle cingulate (139/140)	Postcentral gyrus, medial segment (149/150)	Posterior insula (173/174)

Frontal	Temporal	Occipital	Cingulate	Parietal	Insula
Frontal operculum (119/120)	Inferior temporal gyrus (133/134)	Inferior occipital gyrus (129/130)	Posterior cingulate (167/168)	Parietal operculum (175/176)	
Frontal pole (121/122)	Middle temporal gyrus (155/156)	Lingual gyrus (135/136)		Postcentral gyrus (177/178)	
Gyrus rectus (125/126)	Parahippocampal gyrus (171/172)	Middle occipital gyrus (145/146)		Superior parietal (199/200)	
Lateral orbital gyrus (137/138)	Planum polare (181/182)	Occipital pole (157/158)		Supramarginal gyrus (195/196)	
Medial frontal cortex (141/142)	Planum temporale (185/186)	Occipital fusiform gyrus (161/162)		Precuneus (169/170)	
Middle frontal gyrus (143/144)	Superior temporal gyrus (201/202)	Superior occipital gyrus (197/198)			
Medial orbital gyrus (147/148)	Temporal pole (203/204)				
Postcentral gyrus (149/150)	Transverse temporal gyrus (207/208)				
Precentral gyrus (151/152)					
Superior frontal gyrus (153/154)					
Opercular of inferior frontal gyrus (163/164)					
Orbital part of inferior frontal gyrus (165/166)					
Posterior orbital gyri (179/180)					
Precentral gyrus (183/184)					
Subcalossal area (187/188)					
Superior frontal gyrus (191/192)					
Supplementary motor cortex (193/194)					
Triangular part inferior frontal gyrus (205/206					

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